Personal Digital Vault

1. Project Overview

Personal Digital Vault is a secure web application where registered users can store, organize, and manage their private documents and sensitive credential records in one place. Users can create their own folders and categories, upload files, keep password or account details in a protected form, and manage a simple personal profile. Uploaded files and sensitive data are encrypted, and a file hash is created for each uploaded file to help verify that the stored data has not been changed.

Basic User Flow

This is the only user flow supported by the system.

1. Register or log in
2. Create folders or categories
3. Upload a document or add a secure credential record
4. System validates the item, encrypts sensitive data, and creates a file integrity hash
5. Search, view, download, update, or delete owned items securely

2. Stakeholders

Stakeholder	Responsibility
Project Supervisor / Assessor	Guides the project and evaluates the final work.
Development Team	Designs, builds, and tests the application.
End Users	Store and manage their own documents and credential records.
System Administrator	Manages user accounts and basic application settings.

3. User Roles

Role	Description
User	Registers, logs in, manages a personal profile, creates folders, and uploads and manages their own documents and credential records.
Administrator	Manages user accounts and basic application settings and views a simple usage dashboard. The administrator cannot access users’ decrypted private files.

4. Proposed Technology Stack

Area	Proposed Technology
Front end	JavaScript, HTML, and CSS
Back end	ASP.NET Core Web API using C#
Database	SQL Server Express
Database tool	SQL Server Management Studio (SSMS)
Data access / ORM	Entity Framework Core preferred; ADO.NET as an alternative
Authentication	JWT-based authentication; ASP.NET Core Identity may be used if needed
Password security	Secure password hashing; never store passwords as plain text
File and data security	AES encryption for sensitive files and credential data, and SHA-256 file hashes for integrity checking Protected local server folder outside the public web folder
File storage	

ASP.NET Core,SQL ServerExpress, andSQLServerManagement Studio can allbe used without paid licenses for a university project. SQL Server Express and SSMS are free tools provided by Microsoft.

5. Project Scope

In Scope

- User registration, login, and logout
- Secure user access using JWT
- Personal user profile with basic account details
- Folder and category management
- Uploading, storing, viewing, downloading, renaming, and deleting personal documents
- Adding and managing secure credential / password records
- Encryption of sensitive stored data
- File integrity hash generation
- Search and basic filtering
- Basic administrator account management
- Administrator dashboard showing simple counts such as total users, total uploads, and total stored files

Out of Scope

- Online payments or digital banking
- Public file sharing
- Cloud storage integration
- Mobile application development
- AI document analysis
- Team collaboration features

6. Functional Requirements

Thefollowingaretheonlyfunctionalrequirements for this project.

No.	User Action	System Behaviour
1	User registers and logs in.	System validates the credentials and issues a JWT for secure access.
2	User views and updates their personal profile.	System saves the updated profile details for that user account.
3	User creates, renames, and deletes folders or categories.	System updates the folder structure and links it to the user account.
4	User uploads a document.	System validates the file, encrypts it, stores it in the protected folder, and saves the metadata in the database.
5	User completes a file upload.	System creates and stores a SHA-256 integrity hash for the uploaded file.
6	User adds, views, updates, and deletes secure credential records.	System encrypts sensitive values and masks them by default when displayed.
7	User searches, views, downloads, renames, or deletes items.	System returns and updates only the items owned by that user.
8	User requests a private file or record.	System checks authorization before allowing any access.
9	Administrator manages user accounts.	System allows account management without exposing users' decrypted private content.
10	Administrator opens the dashboard.	System displays simple counts such as total users, total uploads, and total stored files.

7. Non-Functional Requirements

Thefollowingaretheonlynon-functionalrequirements for this project.

Area	**Requirement**
Security	Passwords are hashed, while files and sensitive credential values are encrypted. All access is protected by JWT authentication.
Reliability	The system remains stable during normal use and handles errors without data loss.
Performance	Common actions such as login, search, and file upload respond within a few seconds.
Usability	The interface is simple and clear, so a new user can complete key tasks without training.
Data integrity	SHA-256 hashes are used to confirm that stored files have not been altered.

8. Conclusion

This document defines the agreed business requirements for Personal Digital Vault. The system will give registered users a simple and secure place to store their private documents and credential records, with encryption, hashing, and authorization applied to protect their data.